Lab 9: Explained

Lab 9: Package a trusted set-up as a plugin

The big idea

A plugin is a toolbox packaged for a team: the catalogue lists it, the manifest describes it, a skill adds a workflow, and a hook enforces a safety check.

Marketplace catalogue marketplace.json lists plugins
Plugin manifest plugin.json identifies team-tools
Skill + hook Review code; block prices.csv edits
Install Validate → test locally → install

What you should understand

A plugin packages reusable skills and hooks. Validate it, test it locally, then install it from the marketplace. Hooks can protect sensitive or managed files.

Lab walkthrough — explained in simple English

Goal: share a skill and a hook with your whole team as one installable plugin. Tool: Claude Code · Time: 20 minutes

Before you start

1. Claude Code is installed. In your terminal:

```
claude --version
```

Not installed? Run one of these, then open a new terminal:

• Windows (PowerShell): `irm https://claude.ai/install.ps1 | iex`

• Mac / Linux: `curl -fsSL https://claude.ai/install.sh | bash`

2. Your Claude API key is set in the terminal you will use:

Windows (PowerShell):

```
$env:ANTHROPIC_API_KEY="sk-ant-...your key..."
```

Mac / Linux:

```
export ANTHROPIC_API_KEY="sk-ant-...your key..."
```

3. Python is installed (the plugin's hook is a Python script).

4. First start: the first time you run `claude` in this folder, choose Yes when it asks to use your API key, and Yes when it asks whether you trust this folder.

Mac / Linux: if `python` is not found, change `python` to `python3` in `team-marketplace/plugins/team-tools/hooks/hooks.json`.

Files in this folder

File	What it is
`orders.py`, `billing.py`, `prices.csv`	A tiny shop app to try the plugin on
`team-marketplace/`	A marketplace containing one plugin, `team-tools`

```
team-marketplace/
■■■ .claude-plugin/
■   ■■■ marketplace.json        ← the catalogue (lists the plugins)
■■■ plugins/
    ■■■ team-tools/
        ■■■ .claude-plugin/
        ■   ■■■ plugin.json     ← the manifest (name, version)
        ■■■ skills/
        ■   ■■■ review/SKILL.md ← a review skill (runs in a fork)
        ■■■ hooks/
        ■   ■■■ hooks.json      ← blocks edits to prices.csv
        ■■■ scripts/
            ■■■ protect_files.py
```

Rules to remember:

- Only `plugin.json` goes inside `.claude-plugin/`. Skills and hooks sit at the plugin's root.
- The `name` in `plugin.json` (`team-tools`) becomes the prefix: `/team-tools:review`.
- `hooks.json` uses `${CLAUDE_PLUGIN_ROOT}`, so the script is found wherever the plugin is installed.

Steps

Step 1: Look inside

Open these three files and read them. They are short:

- `team-marketplace/plugins/team-tools/.claude-plugin/plugin.json`
- `team-marketplace/plugins/team-tools/skills/review/SKILL.md`
- `team-marketplace/.claude-plugin/marketplace.json`

Step 2: Validate

In your terminal:

```
cd lab09-plugin
claude plugin validate ./team-marketplace
claude plugin validate ./team-marketplace/plugins/team-tools
```

 `Validation passed` for both.

Step 3: Try the plugin without installing it

```
claude --plugin-dir ./team-marketplace/plugins/team-tools
```

In Claude Code:

```
/team-tools:review billing.py
```

 A short review (no check on `percent`, floats, and so on). It comes from the plugin's skill.

Switch to accept edits with `Shift+Tab` and ask:

```
Change the Pen price in prices.csv to 150.
```

 Blocked by the plugin's hook. Type `/exit`.

Step 4: Install it from the marketplace

In your terminal:

```
claude plugin marketplace add ./team-marketplace
claude plugin install team-tools@team-marketplace
claude plugin list
```

 `team-tools@team-marketplace` is installed. Start `claude` in any folder: `/team-tools:review` is available.

To share it with your team, put `team-marketplace` in a git repository. Each person then runs `claude plugin marketplace add <owner>/<repo>` and installs it the same way.

What you learned

- A plugin bundles skills, hooks and more, with a `plugin.json` manifest.
- A marketplace lists plugins, so a team can install the same tested set-up.
- Use `claude plugin validate` to check it, `--plugin-dir` to test it, and `install` to use it.

Clean up

```
claude plugin uninstall team-tools@team-marketplace
claude plugin marketplace remove team-marketplace
```

REMEMBER

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